Video Style Transfer Evaluation - Synthetic Workflow Demonstration

Claim status: Exploratory
Date: 2026-11-08
Claim: Demonstrate a reproducible style-content-motion evaluation workflow without presenting synthetic clips as model benchmark evidence.

Executive verdict

SSIM ↑: Simulated balanced 0.9290 vs Simulated flicker 0.8111 (better; n=3).

LAB palette W1 ↓: Simulated balanced 9.0208 vs Simulated flicker 7.4201 (worse; n=3).

OFFSET 1 WARP RMSE ↓: Simulated balanced 31.70 vs Simulated flicker 47.77 (better; n=3).

OFFSET 1 FLOW EPE ↓: Simulated balanced 0.2438 vs Simulated flicker 0.5426 (better; n=3).

Runtime ↓: Simulated balanced 18.80 vs Simulated flicker 15.00 (worse; n=3).

Blinded VLM review: Simulated balanced has the highest mean overall visual quality (4.00/5 across 1 reviewer(s)); this is perceptual evidence, not benchmark ground truth.

Most frequent visual failure tag: detail loss (3 occurrence(s)).

The diagnostic style metric measures palette alignment only; texture and brushwork claims require perceptual evidence.

Evidence gates

Gate	Value	Status
Paired valid cases	3	Pass
Methods	3	Pass
Blinded VLM review	Present	Pass
Paper-grade perceptual metrics	Not detected	Not established
Claim status	Exploratory	Caveat

LAB palette distance and Farneback flow are deterministic diagnostic proxies. They are not RAFT, LPIPS, CLIP, FID, or ArtFID.

Quantitative results

Style

Metric	Simulated balanced	Simulated flicker	Simulated overlay
LAB palette W1 (lower)	9.0208 ± 0.5445	**7.4201 ± 0.5347**	<u>8.0416 ± 0.9411</u>
LAB chroma palette W1 (lower)	<u>3.1765 ± 1.3380</u>	**1.8824 ± 1.0178**	4.6879 ± 0.4238

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Content

Metric	Simulated balanced	Simulated flicker	Simulated overlay
SSIM (higher)	**0.9290 ± 0.0165**	0.8111 ± 0.0326	0.7153 ± 0.0149
PSNR (higher)	**22.58 ± 0.9018**	17.54 ± 0.3337	14.90 ± 0.3931

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Temporal and motion

Metric	Simulated balanced	Simulated flicker	Simulated overlay
OFFSET 1 WARP RMSE (lower)	**31.70 ± 6.7038**	47.77 ± 10.70	63.87 ± 6.8744
OFFSET 5 WARP RMSE (lower)	**33.20 ± 5.2688**	46.18 ± 10.55	63.70 ± 7.5960
OFFSET 1 FLOW EPE (lower)	0.2438 ± 0.0558	0.5426 ± 0.3239	**0.1874 ± 0.0547**
OFFSET 1 FLOW CORRELATION (higher)	**0.4268 ± 0.2381**	0.2267 ± 0.0923	0.2662 ± 0.1339
Detrended luminance flicker (lower)	0.1035 ± 0.0619	12.43 ± 0.4689	**0.0822 ± 0.0217**

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Cost

Metric	Simulated balanced	Simulated flicker	Simulated overlay
Runtime (lower)	18.80 ± 0.4000	**15.00 ± 0.4000**	16.50 ± 0.4000
Peak GPU memory (lower)	12.70 ± 0.0000	**9.8000 ± 0.0000**	10.40 ± 0.0000

Best and second-best mean values are shown in bold and underlined, respectively. Valid clips: balanced: 3; flicker: 3; overlay: 3.

Paired comparison against the primary baseline

Metric	d (positive favors primary)	dz	95% CI	n
SSIM	0.1180	7.1536	—	3
LAB palette W1	-1.6007	-15.05	—	3
OFFSET 1 WARP RMSE	16.08	3.9236	—	3
OFFSET 1 FLOW EPE	0.2988	1.0159	—	3
Runtime	-3.8000	n/a	—	3

Blinded visual review

Method	Style adherence	Content preservation	Temporal stability	Artifact control	Overall visual quality
Simulated balanced	4.00/5	4.67/5	3.67/5	3.33/5	4.00/5
Simulated flicker	4.67/5	3.67/5	2.00/5	2.00/5	2.33/5
Simulated overlay	3.33/5	3.00/5	3.00/5	2.00/5	2.00/5

Failure tags: detail loss (3), texture sticking (3), flicker (3), edge ghosting (2), temporal jump (2), texture drift (1), geometry melting (1)

Reviewer records: 1. Scored case/method records: 9. VLM scores are perceptual evidence, not ground truth.

Analysis and discussion

Style fidelity

The available indicators collectively favor Simulated flicker. Reported diagnostics: LAB palette W1 9.0208 ± 0.5445; LAB chroma palette W1 3.1765 ± 1.3380. The blinded style-adherence score is 4.00/5.

Content preservation

The available indicators collectively favor Simulated balanced. Reported diagnostics: SSIM 0.9290 ± 0.0165; PSNR 22.58 ± 0.9018.

Motion and temporal stability

The available indicators collectively favor Simulated balanced. Reported diagnostics: OFFSET 1 WARP RMSE 31.70 ± 6.7038; OFFSET 5 WARP RMSE 33.20 ± 5.2688; OFFSET 1 FLOW EPE 0.2438 ± 0.0558; Detrended luminance flicker 0.1035 ± 0.0619.

Perceptual review and failure modes

The blinded review gives the highest overall visual quality to Simulated balanced (4.00/5 across 1 reviewer(s)). This reflects preference on the reviewed sample rather than ground-truth quality. The leading failure modes are detail loss (3), texture sticking (3), flicker (3).

Computational cost

The available indicators collectively favor Simulated flicker. The primary method uses runtime 18.80 ± 0.4000 s and peak memory 12.70 ± 0.0000 GB.

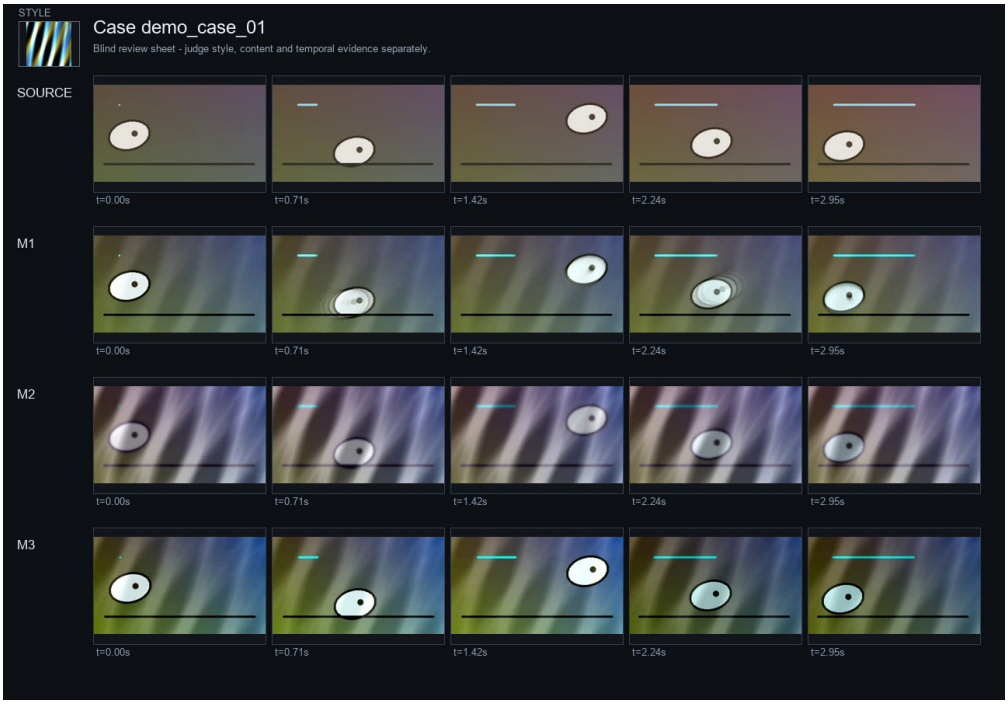
Synthesis

The results show a multidimensional trade-off rather than universal dominance; the current evidence does not support a claim that one method is uniformly superior.

Per-case outcomes

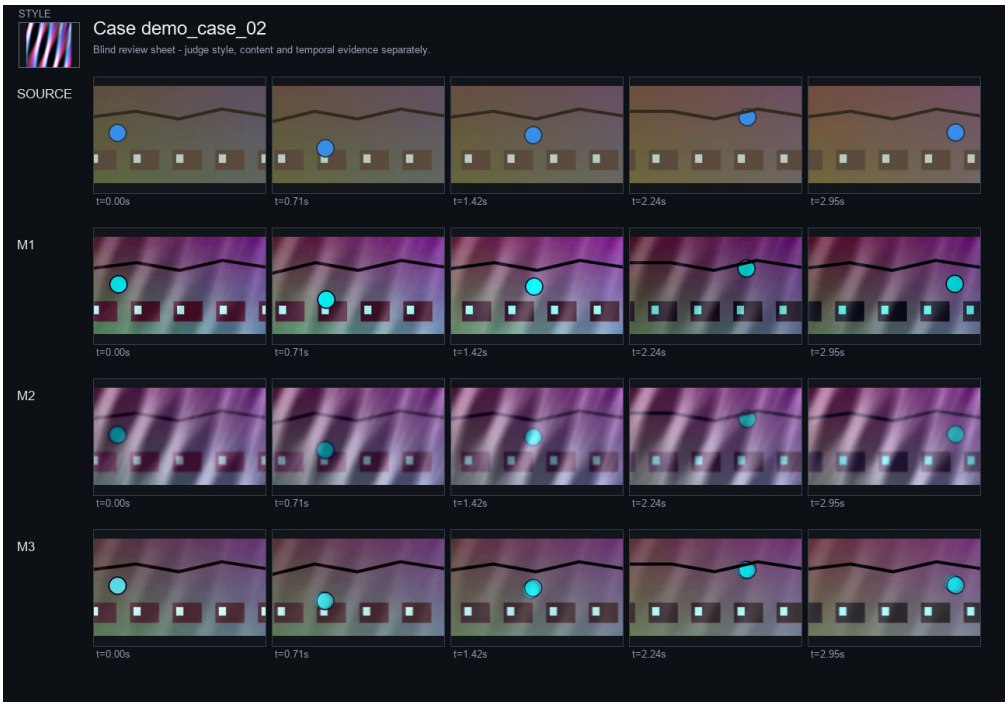
demo_case_01

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9395	8.8136	28.61	0.2963
Simulated flicker	0.8270	7.3344	44.06	0.5494
Simulated overlay	0.7122	8.1708	65.43	0.1389



demo_case_02

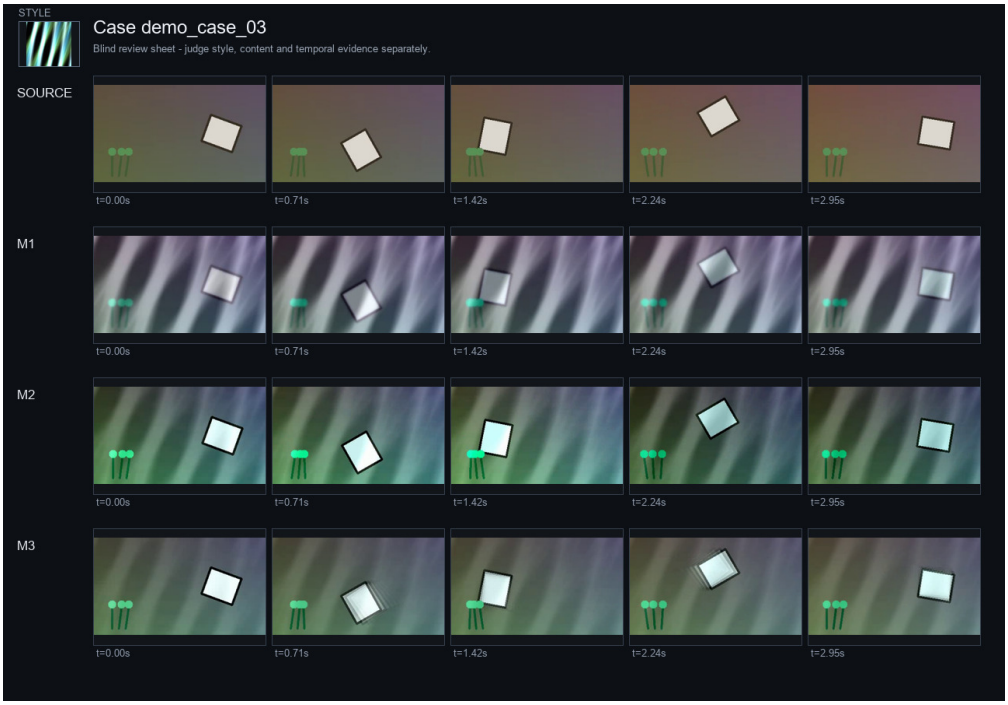
Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9101	8.6103	39.39	0.2500
Simulated flicker	0.7736	6.9334	59.84	0.8630
Simulated overlay	0.7315	7.0425	69.83	0.2467



demo_case_03

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated balanced	0.9376	9.6385	27.09	0.1852

Method	SSIM (higher)	Palette W1 (lower)	Warp RMSE d1 (lower)	Flow EPE d1 (lower)
Simulated flicker	0.8327	7.9925	39.42	0.2154
Simulated overlay	0.7022	8.9114	56.35	0.1766



Threats to validity

- Content and style distributions may not represent the intended use.
- Low-level metrics are sensitive to colorization and compression.
- Farneback flow is not RAFT and can fail on large motion or occlusions.
- A single style image does not define a valid FID style distribution.
- VLM review can have position, content, and prompt bias.
- External baseline numbers copied from papers may not use the same settings.
- Fewer than five clips do not support a stable comparative confidence interval.

Conclusion and recommendation

This report uses 3 synthetic cases to validate the evaluation pipeline and does not provide evidence of real model performance. The tables, perceptual analysis, and conclusions preserve multidimensional evidence and should not be interpreted as a benchmark ranking of EchoStyle or the proposed method.

Recommendation

Use these results to validate the evaluation pipeline and choose the next experiment. Do not treat the synthetic run as model evidence.